Training Day 15

1. Topics Covered and Brief Explanation

1. Introduction to Scikit-learn

Scikit-learn is a popular Python library used for implementing Machine Learning algorithms. It provides tools for data preprocessing, model training, prediction and model evaluation.

2. Features of Scikit-learn

- Provides various machine learning algorithms.
- Supports classification and regression.
- Provides tools for data preprocessing.
- Helps in splitting datasets into training and testing sets.
- Provides functions for evaluating model performance.

3. Training Data and Testing Data

Learned the importance of dividing a dataset into two parts:

- Training Data: Used to teach the model and help it identify patterns.
- Testing Data: Used to check how well the trained model performs on unseen data.

4. Dataset Splitting

Studied the train_test_split() function of Scikit-learn, which divides a dataset into training and testing portions.

5. Model Training and Prediction

Learned the basic process of creating a model, training it using fit() and generating predictions using predict().

2. Tools and Technologies Used

Tool	Purpose
Python	Writing machine learning programs.
Scikit-learn	Model creation, training and testing.
Pandas	Organizing and handling datasets.
NumPy	Numerical operations.

Google Colab Executing Python code.

3. Practical Learning / Code Example

Example: Training a Simple Classification Model

```
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
# Sample dataset
X = [[1], [2], [3], [4], [5], [6], [7], [8], [9], [10]]
y = ["Low", "Low", "Low", "Low", "Low",
     "High", "High", "High", "High", "High"]
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Create model
model = KNeighborsClassifier(n_neighbors=3)
# Train model
model.fit(X_train, y_train)
# Make predictions
predictions = model.predict(X_test)
print("Actual Values:", y_test)
print("Predicted Values:", predictions)
```

Explanation:

- train_test_split() divides the dataset into training and testing sets.
- test_size=0.2 reserves 20% of the data for testing.
- KNeighborsClassifier() creates the ML model.
- fit() trains the model.
- predict() generates predictions for the testing data.

4. Learning Outcomes

- Understood the purpose and features of Scikit-learn.
- Learned the importance of training and testing datasets.
- Gained knowledge of splitting datasets using Python.
- Understood the basic steps involved in model training.
- Practised creating a classification model and generating predictions.